



IILM UNIVERSITY
Greater Noida, UTTAR PRADESH

ACADEMIC LEARNING REPORT

AICTE VIRTUAL INTERNSHIP PROGRAMME

ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

An Internship Presentation Report on Artificial Intelligence, Machine Learning & Deep Learning

Internship Organization Kodacy

Domain Artificial Intelligence & Machine Learning

Presented By Arvinder Singh Bagga
Roll No.:2410031176
Section: 3CSE24

Completed on:- 8th September, 2026

Candidate & Internship Overview

Summary of candidate profile, internship engagement details, and core technical learning tracks.



Candidate & Track Details

Candidate Name: Arvinder Singh Bagga

Roll No.: 2410031176

Section: 3CSE24

Host Organization: Kodacy

Domain: AI & Machine Learning

Internship Mode: AICTE Virtual Mode

Institution: IILM University, Greater Noida



Internship Focus Areas

- ✓ Artificial Intelligence & Problem-Solving Concepts
- ✓ Machine Learning Classifiers & Regression Models
- ✓ Deep Learning Fundamentals & Multi-Layer Networks
- ✓ Convolutional & Recurrent Neural Architectures (CNN & RNN)
- ✓ Python Programming Foundation for Data & AI
- ✓ End-to-End Machine Learning Model Development Workflow

Internship Profile

Structured learning framework provided by Kodacy for mastering AI & ML concepts.



Structured Learning Experience

The internship provided a comprehensive and structured curriculum in the field of Artificial Intelligence and Machine Learning. The programme focused on developing foundational and conceptual understanding of intelligent systems, statistical decision making, algorithmic models, neural networks, and Python-driven workflows.

Through guided modules, interactive study tracks, and conceptual case reviews, the curriculum bridged theoretical computer science principles with modern data-driven intelligence methods.



Artificial Intelligence

Intelligent Systems



Machine Learning

Supervised & Unsupervised



Deep Learning

Neural Networks



Python Foundation

Programming & Syntax



Modern AI & Applications

Chatbots & Conversational AI Concepts

Core Learning Objectives

The primary technical pillars and learning outcomes established for the internship program.



1. AI Fundamentals

Studying the core principles, historical evolution, intelligence classification, and real-world application paradigms of Artificial Intelligence.



2. Machine Learning

Understanding how machines identify patterns from structured data, make probabilistic estimates, and optimize predictive models.



3. Deep Learning

Exploring multi-layered artificial neural networks, activation functions, gradient flow, and specialized CNN & RNN structures.



4. Model Workflow

Learning the foundational steps of data handling, model training, evaluation metrics, and hyperparameter tuning in Python.

Artificial Intelligence Fundamentals

Foundational concepts and primary specialized branches explored within Artificial Intelligence.



Core Concepts

- ✓ Introduction to Intelligent Agents
- ✓ Key Challenges & Limitations in AI
- ✓ Reasoning and Problem-Solving
- ✓ Narrow vs. General Intelligence (AGI)



Language & Vision

- ✓ Natural Language Processing (NLP)
- ✓ Computer Vision Concepts
- ✓ Feature Recognition in Images
- ✓ Speech & Audio Processing Basics

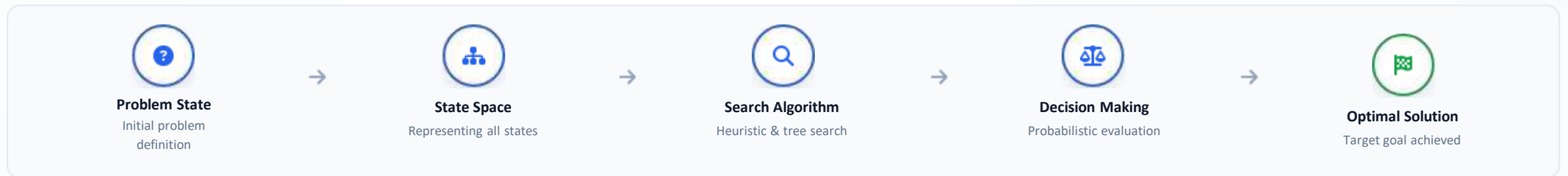


Cognition & Systems

- ✓ Cognitive Computing Paradigms
- ✓ Knowledge Representation Schemes
- ✓ Expert Systems & Inference Engines
- ✓ Ethical & Responsible AI Basics

AI Problem Solving & Intelligent Systems

State space search representations, probabilistic decision frameworks, and algorithmic solutions.



Search & Game Theory

- ✓ State Space Representation & Graph Traversal
- ✓ Minimax Algorithm for Two-Player Adversarial Games
- ✓ Evolutionary Computation & Genetic Algorithms

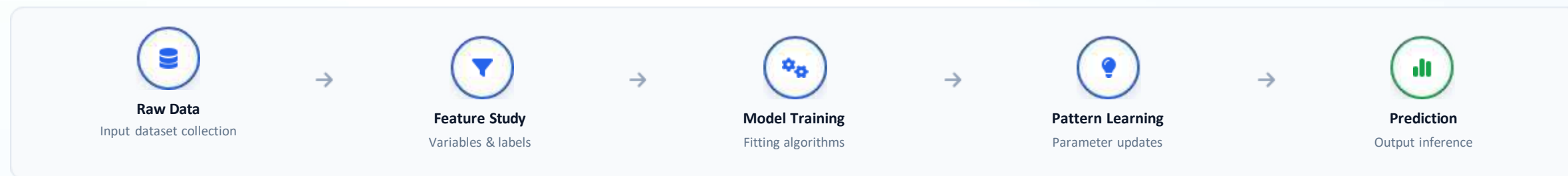


Probability & Decision Making

- ✓ Probability Fundamentals & Conditional Likelihood
- ✓ Bayes' Theorem & Posterior Probability Calculation
- ✓ Naive Bayes Classifier for Text & Categorical Data

Machine Learning Concepts

Fundamental learning paradigms, supervised algorithms, and model training concepts.



Classification & Regression

- ✓ Supervised vs Unsupervised vs Reinforcement Learning
- ✓ K-Nearest Neighbours (KNN) Distance-Based Classifier
- ✓ Linear Regression for Continuous Target Variables
- ✓ Logistic Regression & Sigmoid Curve for Binary Outcomes



Ensemble Methods

- ✓ Decision Tree Classification & Information Gain (Entropy/Gini)
- ✓ Random Forests: Bagging & Reducing Model Variance
- ✓ Gradient Boosting Concepts for Sequential Error Correction
- ✓ Introduction to XGBoost for Tabular Data Performance

Deep Learning & Neural Networks

Biological inspirations, multi-layer artificial neurons, and advanced neural architectures.



Artificial Neural Network Anatomy

- ✓ **Perceptron Unit:** Inputs, Synaptic Weights, Biases, and Summation
- ✓ **Activation Functions:** ReLU, Sigmoid, and Tanh Non-Linearities
- ✓ **Multi-Layer Structure:** Input Layer → Hidden Layers → Output Layer
- ✓ **Training Concept:** Forward Propagation, Loss Computation & Backpropagation



Specialized Neural Architectures

- ✓ **CNN:** Convolution & Pooling for Spatial Feature Extraction
- ✓ **RNN:** Recurrent Feedback Loops for Sequential & Temporal Dependencies
- ✓ **GANs:** Generator and Discriminator Adversarial Learning Concept

INPUT LAYER

Feature Vectors



HIDDEN LAYERS

Weights & Activations



OUTPUT LAYER

Probabilities / Classes

CNN (Vision)

RNN (Sequence)

GAN (Generative)

Technologies & Tools Used

Tools and technologies explored throughout the AI & Machine Learning internship.

Programming & Data Tools

- **Python**
Programming foundation for AI/ML workflows
- **Data Structures**
Lists, tuples, sets and dictionaries
- **Data Handling**
Feature exploration and preparation
- **Scikit-Learn**
Model building and evaluation pipelines

AI & Machine Learning Tools

- **Machine Learning**
Classification, regression and ensembles
- **Deep Learning**
ANN, CNN and RNN concepts
- **XGBoost**
Gradient boosting and parameter tuning
- **Modern AI**
Chatbots, ChatGPT and LLM concepts

Integrated Learning Python + data preparation + ML models + neural networks + responsible modern AI

Python Programming for AI & ML

Foundational programming constructs and data structures studied for AI/ML problem solving.

Python Basics & Structures

- ✓ **Primitive Data Types:** Integers, Floats, Strings, Booleans
- ✓ **Data Collections:** Lists (mutable), Tuples (immutable), Sets (unique)
- ✓ **Dictionaries:** Key-value paired storage for structured parameters

Logic & Functionality

- ✓ **Control Flow:** Conditional Statements (if/elif/else)
- ✓ **Iterative Loops:** for and while loops for data scanning
- ✓ **Functions:** Modular functions and anonymous lambda expressions



Python
Fundamentals



Data
Types



Data
Structures



Conditional
Statements



Loops &
Iterations



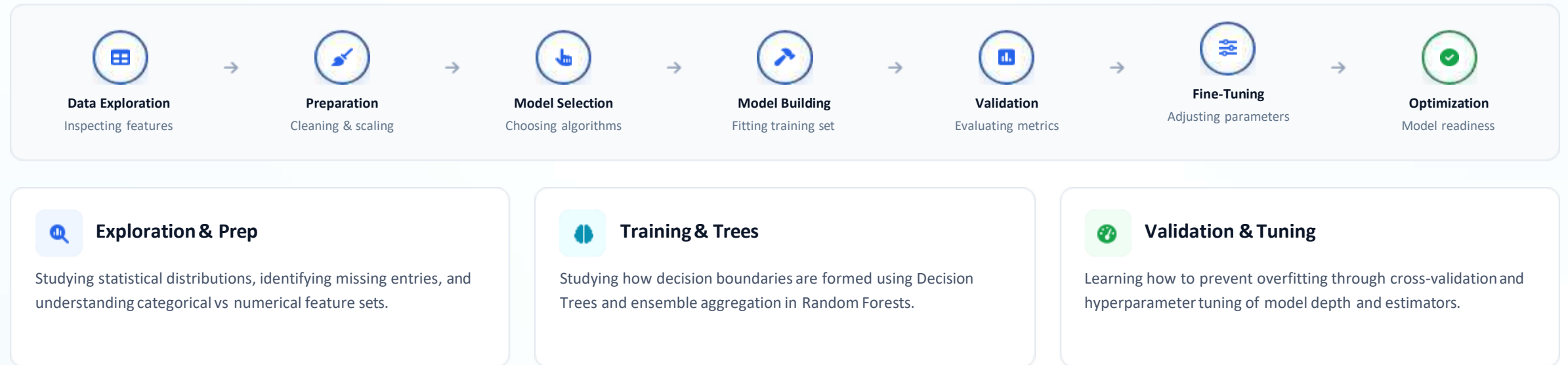
Modular
Functions



AI & ML
Programming

Machine Learning Model Development

Understanding the end-to-end lifecycle and stages involved in building predictive models.



Advanced Machine Learning Concepts

Data pipeline integrity, model validation rigor, and boosting mechanisms.



Data Handling

- ✓ Missing value imputation
- ✓ One-Hot & Label Encoding
- ✓ Scikit-Learn Pipelines



Evaluation Rigor

- ✓ K-Fold Cross-Validation
- ✓ Precision, Recall & F1-Score
- ✓ ROC-AUC Curve Analysis



Gradient Boosting

- ✓ Residual learning concept
- ✓ Parameter tuning strategies
- ✓ XGBoost efficiency concepts



Model Reliability

- ✓ Preventing Data Leakage
- ✓ Train-Test Contamination
- ✓ Generalization testing

Applied & Modern Artificial Intelligence

Conversational systems, foundational Generative AI principles, and real-world implications.



Chatbot Development

- ✓ Rule-based vs Intent-driven systems
- ✓ Utterance classification & entity parsing
- ✓ Dialogue management flowcharts



Understanding ChatGPT

- ✓ Concepts of Large Language Models (LLMs)
- ✓ Prompt structuring & conversational AI
- ✓ Generative text synthesis basics



Present & Future AI

- ✓ Applications across healthcare & industry
- ✓ Intelligent automation & assistive AI
- ✓ Ethical governance & AI safety

Skills & Knowledge Developed

Summary of technical understanding and analytical skills acquired across the internship.



Artificial Intelligence

Understanding intelligent systems, search algorithms, state spaces, and knowledge representations.



Machine Learning

Understanding classification, regression, Bayes rule, Decision Trees, and Random Forests.



Deep Learning

Knowledge of multi-layer perceptrons, activation functions, CNN vision, and RNN sequential models.



Python Foundation

Developing essential programming logic, data structure manipulation, and functional methods.



ML Workflow

Understanding data cleaning, feature exploration, cross-validation, and hyperparameter tuning.



Modern AI & LLMs

Exposure to conversational agents, ChatGPT architecture concepts, and responsible AI practices.

My AI & ML Learning Journey

Sequential curriculum progression traversed throughout the internship tenure at Kodacy.

STEP 01**AI Fundamentals**

History, core branches & intelligent agents

STEP 02**Problem Solving**

Search trees, heuristics & state spaces

STEP 03**Probability & Bayes**

Statistical decision making & Bayes theorem

STEP 04**Machine Learning**

Supervised classification & regression models

STEP 05**Neural Networks**

ANN, Perceptrons & backpropagation

STEP 06**CNN & RNN**

Computer vision & sequential NLP architectures

STEP 07**Python Foundation**

Data structures, loops & function logic

STEP 08**Model Development**

Data cleaning, training & validation pipeline

STEP 09**Advanced ML**

Gradient Boosting, tuning & data leakage prevention

STEP 10**Modern AI & LLMs**

Chatbots, ChatGPT & future AI horizons

My 4-Week AI & ML Learning Journey

A structured progression from foundational concepts to applied artificial intelligence.

WEEK 01

1

AI & ML Foundations

Artificial Intelligence concepts, intelligent agents, problem-solving, search algorithms, probability and Bayes' theorem.

WEEK 02

2

Machine Learning

Supervised and unsupervised learning, KNN, regression, Decision Trees, Random Forests and ensemble methods.

WEEK 03

3

Deep Learning & Python

Neural networks, activation functions, backpropagation, CNN and RNN architectures, plus Python programming foundations.

WEEK 04

4

Model Development & Modern AI

Data preparation, training, validation, hyperparameter tuning, advanced ML practices, chatbots and LLM concepts.

Internship Experience & Conclusion

Reflective summary of academic growth, practical insights, and future learning pathways.



Key Internship Reflections

The Artificial Intelligence and Machine Learning virtual internship provided a well-structured and disciplined learning experience covering both theoretical foundations and practical algorithmic workflows.

Throughout the program, I explored intelligent problem solving, statistical decision-making, supervised and unsupervised machine learning algorithms, deep neural architectures, and end-to-end model evaluation techniques in Python.

Academic Value: Established a strong technical foundation for advanced academic coursework and future machine learning projects.



Future Learning Pathways

- **Hands-On Deep Learning Projects**
Building custom computer vision and NLP models using PyTorch & TensorFlow.
- **Advanced Large Language Models**
Exploring fine-tuning techniques, prompt engineering, and RAG architectures.
- **MLOps & Deployment Workflows**
Understanding cloud-based API serving, model tracking, and continuous monitoring.

Formal certificate of completion awarded by Kodacy under the AICTE Virtual Internship Programme.



Credential Details

Awarded To: **Arvinder Singh Bagga**

Organization: **Kodacy**

Domain: **AI & Machine Learning**

Program: **AICTE Virtual Internship**

Status: **Successfully Completed**



Kodacy

SPACE
Scientific Platforms And Cosmic Explorations



Certificate

OF INTERNSHIP

This certificate is presented to :

Arvinder Singh Bagga

for successfully completing the 30 day virtual internship programme on
Artificial Intelligence and Machine Learning
conducted by KODACY in association with Scientific
Platforms And Cosmic Explorations (SPACE).



Date Of Completion : September 8, 2025
Certificate ID : 544b2875e2f9e315
◀ Scan this QR to check validity

Authorized Signature
Akash Joseph (CEO)



Skills & Capabilities Gained

Key technical, analytical and professional capabilities developed during the internship.

AI & Problem Solving

Understand intelligent agents, state spaces, search strategies, reasoning and probabilistic decision-making.

Machine Learning

Apply classification, regression, Bayes, Decision Trees, Random Forests and ensemble learning concepts.

Deep Learning

Explain neural networks, activation functions, backpropagation, CNNs, RNNs and modern architectures.

Python & Data Handling

Use Python fundamentals, data structures, functions, data preparation and model-oriented workflows.

Model Development

Perform training, validation, evaluation, cross-validation, tuning and data-leakage prevention.

Modern AI Awareness

Understand chatbots, LLM concepts, conversational AI, applications and responsible AI practices.



Thank You!

Thank you for your time and guidance. Open to questions and academic evaluation.

Arvinder Singh Bagga

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IILM University, Greater Noida